AITS
ALL INDIA TEST SERIES

Dropper JEE Advanced

AJPM9.1/08 **Test- 08**

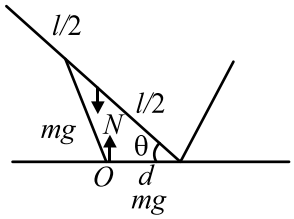
DURATION : 180 Minutes **DATE : 16/02/2025** **M.MARKS : 180**

ANSWER KEY

PHYSICS		CHEMISTRY		MATHEMATICS	
1.	(0.80)	19.	(9.48)	37.	(7.00)
2.	(0.82)	20.	(1.20)	38.	(13.75)
3.	(1.27)	21.	(6.00)	39.	(5.20)
4.	(4.00)	22.	(1.00)	40.	(0.00)
5.	(8.00)	23.	(54.00)	41.	(47.00)
6.	(100.00)	24.	(0.48)	42.	(164.00)
7.	(3.00)	25.	(4.00)	43.	(6.00)
8.	(35.00)	26.	(2.39)	44.	(1.50)
9.	(A, B, C, D)	27.	(A, B, D)	45.	(B, C)
10.	(A, C)	28.	(B, C, D)	46.	(B, D)
11.	(A, C)	29.	(A, B, C, D)	47.	(B, C)
12.	(B, C)	30.	(A, B, C)	48.	(C, D)
13.	(A, C)	31.	(A, B, C)	49.	(A, D)
14.	(A, B)	32.	(A, B)	50.	(A, B, D)
15.	(B)	33.	(A)	51.	(C)
16.	(C)	34.	(A)	52.	(A)
17.	(C)	35.	(B)	53.	(D)
18.	(B)	36.	(D)	54.	(D)

PART-I (PHYSICS)

1. (0.80)

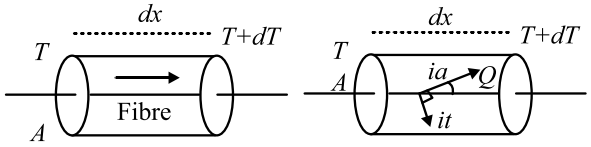


When it is about to topple, N will pass through *O*

$$Mg\left(\frac{l}{2}\cos\theta - d\right) = mg\frac{d}{2}$$

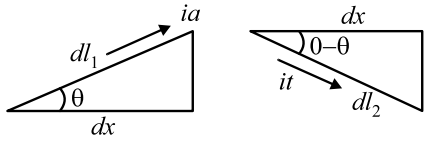
$$\Rightarrow l = 0.8m$$

2. (0.82)



$$\text{Heat flux} \rightarrow \left(\frac{dQ}{dt}\right)_1 = i_1 = -\eta KA\frac{dT}{dx} - (i)$$

$$\left(\frac{dQ}{dt}\right)_2 = i_2 = i_a + i_t$$



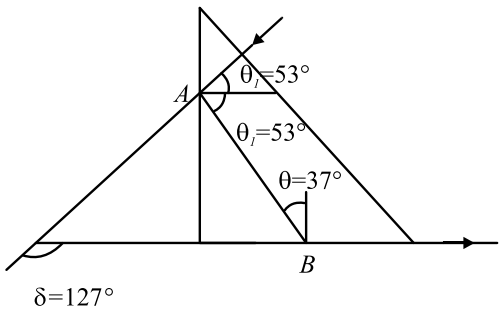
$$i_a = \eta k(A\cos\theta)\frac{dT}{dl_1} = \eta kA[\cos\theta]^2\frac{dT}{dx}$$

$$i_t = k(A\sin\theta)\frac{dT}{dl_2} = kA\left[\sin^2\theta\right]\frac{dT}{dx}$$

$$\Rightarrow \frac{i_2}{i_1} = \frac{\eta\cos^2\theta + \sin^2\theta}{\eta} = 0.82$$

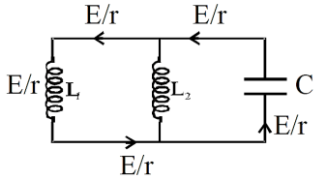
3. (1.27)

At point *A* μsin θ > 1 Hence total internal reflection will take place At point *B* μ sin θ_1= 1. Therefore, light ray will graze the surface hence δ =127°

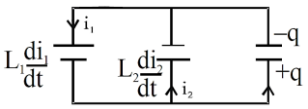


4. (4.00)

t = 0

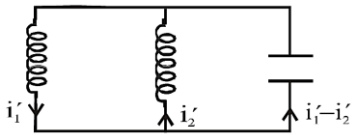


general *t* = *t*



So, $L_1\frac{di_1}{dt} = -L_2\frac{di_2}{dt} = \frac{-q}{C}$

for maximum current in L_2.



$$\Rightarrow L_1\left(i_1' - \frac{E}{r}\right) = -L_2\left(i_2' - 0\right) \qquad \ldots(i)$$

$$\frac{1}{2}L_1\left(\frac{E}{r}\right)^2 = \frac{1}{2}L_1 i_1'^2 + \frac{1}{2}L_2 i_2'^2 \qquad \ldots(ii)$$

$$\Rightarrow \frac{L_1E^2}{r^2} = L_2 i_2'^2 + L_1\left(\frac{E}{r} - \frac{L_2 i_2'^2}{L_1}\right)^2$$

$$\Rightarrow i_2' = \frac{2EL_1}{r\left(L_1 + L_2\right)}$$

E = 5V

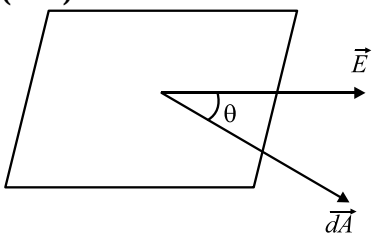
L_1 = 4 mH

L_2 = 1 mH

$r = 2\Omega$

$$\Rightarrow i_2' = \frac{2\times5\times4}{5\times2} = 4\,\text{A}$$

5. (8.00)



Let charge on each surface $q = \sigma l^2$

Flux due to other three surfaces, $\phi = \frac{q}{2\varepsilon_0}$

Force, F = $\int dF\cos\theta = \int dqE\cos\theta = \int \sigma dAE\cos\theta$

$$F = \sigma\int \vec{E}\cdot d\vec{A}$$

$$= \sigma\times\frac{q}{2\varepsilon_0}$$

$$= \sigma\times\frac{\sigma A}{2\varepsilon_0}$$

$= \dfrac{\sqrt{3}\sigma^2 l^2}{8\varepsilon_0}$.

$x = \dfrac{8}{\sqrt{3}}$

$\sqrt{3}x = 8.$

6. (100.00)

$P = \dfrac{1}{2}\mu\omega^2 \cdot a^2 v$ Now $\omega = \dfrac{2\pi v}{\lambda}$ and $v = \sqrt{\dfrac{T}{\mu}}$

Therefore, $P = \dfrac{2\pi^2 a^2}{\lambda^2}\sqrt{\dfrac{T^3}{\mu}}$

Substituting, $a = 10\,\text{mm}$, $\lambda = 0.5\,\text{m}$,
$T = 500$ N and $\mu = 0.8$ kg/m
We get $P = 100\,\text{W}$

7. (3.00)

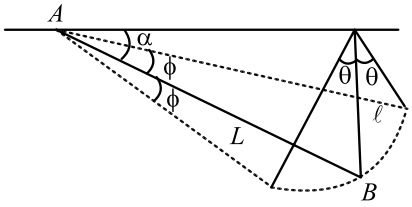
Suppose the length of the rod is L and the mass of the rod is *m*
In equilibrium, tension in the thread

$F = \dfrac{mg}{2}$ …(i)

From geometry,
and $l\theta = \text{L}\phi$ …(ii)

Now, equation of motion is

$\left(\dfrac{mL^2}{3}\right)\dfrac{d^2\phi}{dt^2} = -(F\sin\theta)L$



$= -\text{FL}\theta \quad (\because \sin\theta \approx \theta)$

$= -\left(\dfrac{mg}{2}\right)\text{L}\left(\dfrac{\text{L}\phi}{l}\right)$

(Using (i) and (ii))

$= -\left(\dfrac{\text{mgL}^2}{2l}\right)\phi \;\Rightarrow\; \dfrac{d^2\phi}{dt^2} = -\left(\dfrac{3g}{2l}\right)\phi$

$\Rightarrow\ \omega = \sqrt{\dfrac{3\text{g}}{2l}}$

Thus, $T = \dfrac{2\pi}{\omega} \Rightarrow\ T = 2\pi\sqrt{\dfrac{2l}{3\text{g}}}$

8. (35.00)

$\dfrac{1}{Z} = \dfrac{1}{R + jX_L} - \dfrac{1}{jX_C} = \dfrac{R}{R^2 + X_L^2} + j\left(\dfrac{1}{X_C} - \dfrac{X_L}{R^2 + X_L^2}\right)$

For the resonance, the imaginary part should be absent.

$\therefore\quad \dfrac{1}{X_C} - \dfrac{X_L}{R^2 + X_L^2} = 0$

$\Rightarrow\quad R^2 + X_L^2 - X_C X_L = 0$

$\Rightarrow\quad R^2 + (\omega L)^2 - \left(\dfrac{1}{\omega C}\right)(\omega L) = 0$

$\Rightarrow\quad \omega = \sqrt{\dfrac{1}{LC} - \dfrac{R^2}{L^2}}$

$\Rightarrow\quad v = \dfrac{1}{2\pi}\sqrt{\dfrac{1}{LC} - \dfrac{R^2}{L^2}}$

9. (A, B, C, D)

$\dfrac{1}{2}mv^2 + Q\left(\dfrac{P\cos\theta}{4\pi\varepsilon_0 r^2}\right) = 0$

Or $v = \sqrt{\dfrac{-2QP\cos\theta}{4\pi\varepsilon_0 mr^2}}\left\{\dfrac{\pi}{2} \le \theta \le \pi\right\}$

Circular motion of bead requires a centripetal force $E_r = -\dfrac{\partial V}{\partial r} = \dfrac{2P\cos\theta}{4\pi\varepsilon_n mr^3}$,

Note that $QE_r = \dfrac{mv^2}{r}$

10. (A, C)

$\theta = 0, F = mg.\mu_s = \tan\theta.$

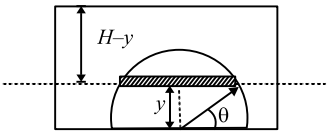
at $\theta = 60, F = mg\cos\theta\sqrt{1+\mu^2}$

11. (A, C)

$dF = pdA = \rho g(H - R\sin\theta)\cdot 2R\cos\theta \cdot R\cos\theta d\theta$

$F = 2R^2\rho g\int_0^{\pi/2}(H - R\sin\theta)\cos^2\theta d\theta$

$= 2R^2\rho g\left(\dfrac{H\pi}{4} - \dfrac{R}{3}\right)$



$d\tau = dFy \quad \tau = 2R^3\rho g\int_0^{\pi/2}(H - R\sin\theta)\cos^2\theta\sin\theta d\theta$

$= 2R^3\rho g\left(\dfrac{H}{3} - \dfrac{\pi R}{16}\right)$

12. (B, C)

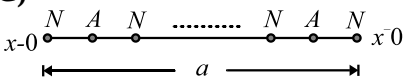


Figure shows the altermate nodes and antinodes in the region between $x = 0$ and $x = a$. In the nth mode, there are *n* nodal points. The wavelength, λn, of the stationary waves, formed is, therefore, given by

$\left(\dfrac{\lambda_a}{2}\right) = a; \quad \therefore \lambda_n = \dfrac{2a}{n}$

Let P_n be the linear momentum of the particle. We are given that

$\lambda_n = \dfrac{h}{p_n} = \dfrac{h}{\sqrt{2mE_n}} \therefore \quad \dfrac{2a}{n} = \dfrac{h}{\sqrt{2mE_n}} \therefore \quad E_n = \dfrac{n^2h^2}{8ma^2}$

For a given value of *m*; $E_n \propto a^{-2}$

Also, $mv_n \propto p_n = \frac{hn}{2a} \quad \therefore v_n = \frac{nh}{2ma} \quad \therefore \quad v_n \propto n$

13. (A, C)

$C_{eq} = C/3$

$Q_0 = CV_0/3$

$\omega = \sqrt{\frac{3}{LC}}$

$q = Q_0 \cos(\omega t)$

$t_0 = 2\pi\sqrt{\frac{LC}{3}}$

$q = Q_0$

$\Delta U_{loss} = \frac{Q_0^2}{2\left(\frac{C}{3}\right)} - \frac{Q_0^2}{2\left(\frac{C}{2}\right)} = \frac{Q_0^2}{2C}(3-2) = \frac{Q_0^2}{2C}$

$\Delta U_{loss} = \frac{CV_0^2}{18}$

If $t_0 = \frac{\pi}{3}\sqrt{\frac{LC}{3}} \quad q = Q_0 \cos\left(\frac{\pi}{3}\right)$

$q = \frac{Q_0}{2}$

$\left(U_f\right)_{capacitor} = \frac{\left(\frac{Q_0}{2}\right)^2}{2\left(\frac{C}{2}\right)} = \frac{Q_0^2}{4C}$

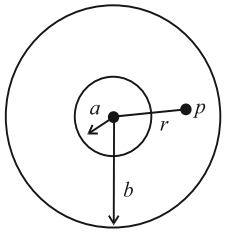
$\left(U_f\right)_{inductor} = \frac{Q_0^2}{2\left(\frac{C}{3}\right)} = \frac{\left(\frac{Q_0}{2}\right)^2}{2\left(\frac{C}{3}\right)} = \frac{9Q_0^2}{8C}.$

$\left(U_f\right)_{Net} = \frac{11Q_0^2}{8C} = \frac{Q_{max}^2}{2\left(\frac{C}{2}\right)}$

$Q_{max} = \sqrt{\frac{11}{8}}\left(\frac{CV_0}{3}\right)$

$= \sqrt{\frac{11}{2}}\left(\frac{CV_0}{6}\right)$

14. (A, B)



$\int dR = \int_a^b \rho \frac{dr}{4\pi r^2}$

$R = \frac{\rho(b-a)}{4\pi ab}$

$i = \frac{V}{R}$

$J = \frac{i}{4\pi r^2} = \frac{Vab}{\rho(b-a)r^2}$

$J = \frac{E}{\rho} \Rightarrow E = \frac{Vab}{(b-a)r^2}$

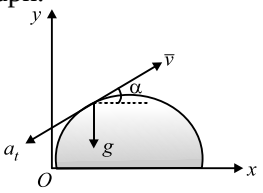
15. (B)

Path equation is $y = Ax - Bx^2$

$S = \frac{dy}{dx} = A - 2Bx = A - 2B(u_x t)$

$S = A-(2Bu_x t)$

This is in the form of $y=-mx+C$. Hence S is the suitable graph.



(B) As the particle rises from point of projection O to highest point, α decrease from θ to $O \Rightarrow$ a_t decreases from gsin θ to zero. Θ = angle of projection. After that, a_t again increase from O to gsin θ.

This variation is best represented by (R)

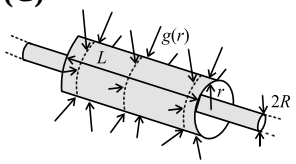
(C) Normal acceleration a_N = gcos α·a_N increases from gcos θ to g and then decrease to gcos θ again. Thin is best represented by Q.

(D) $v = \sqrt{v_x^2 + v_y^2} = \sqrt{(u\cos\theta)^2 + (u\sin\theta - gt)^2}$

$v = \sqrt{u^2 + g^2t^2 - 2u(\sin\theta)gt}$

With increase in t, v first decreases and then increases after reaching maximum height. But $v_{minimum} \neq 0$. This is best represented by P.

16. (C)



Accordingly, for a coaxial cylinder of length L and radius r ($r > R$) that surrounds the planet, we have

$g \cdot 2\pi rL = 4\pi G \cdot \pi R^2 L\rho$, that is $g(r) = \frac{2\pi G\rho R^2}{r}.$

Here, ρ denotes the average density of the planet, which is assumed to have a cylindrically symmetric mass distribution.

Using the form of the gravitational field strength, the velocity can be found. The condition for the orbital motion of a point-like body with mass m around a circular trajectory of radius r is

$mg(r) = m\frac{v^2}{r},$

$v = R\sqrt{2\pi G\rho} = 9.8\,\mathrm{km\,s^{-1}}.$

The period of a satellite orbiting in a circle of radius r is $2\pi r/v_{c,\,1}$, so if the rotational period of

the planet around its axis is T_0 = 1 day, then the radius of the trajectory of a 'geostationary' communications satellite is

$$r_0 = \frac{T_0 v}{2\pi} = R\sqrt{\frac{T_0^2 G\rho}{2\pi}} = 1.33\times10^8 \text{ m.}$$

The relationship between the two distances can be written in the form

$$r_0 = \sqrt{\frac{3\left(r_0^{\text{Earth}}\right)^3}{2R}}$$

17. (C)

$\Rightarrow$ $(S_2P + (\mu - 1)t_2) - (S_1P + (\mu - 1)t_1) = n\lambda$

Bright fringe

$\Rightarrow$ $d\sin(\theta) = n\lambda - (\mu - 1)(t_2 - t_1)$

$$\Rightarrow \quad \sin(\theta) = \frac{-(\mu-1)(t_2-t_1)}{d} = \frac{-2}{3}$$

$$\tan(\theta) = \frac{2}{\sqrt{5}}$$

$$x = \frac{6}{\sqrt{5}}$$

$$\Rightarrow \quad -1 \le \left\{\sin(\theta) = \frac{n\lambda - (\mu-1)(t_2-t_1)}{d}\right\} \le 1$$

$$\Rightarrow \quad \frac{(\mu-1)(t_2-t_1)-d}{\lambda} \le n \le \frac{(\mu-1)(t_2-t_1)+d}{\lambda}$$

$-5.5 \le n \le 27.7$

33 number of fringes observed

If film is displaced by d then

$$\frac{(\mu-1)\frac{5t}{3}-d}{\lambda} \le n \le \frac{(\mu-1)\frac{5t}{3}+d}{\lambda}$$

$11.1 \le n \le 44.4$

33 fringes observed.

If film is displaced by 1.5 d then

$$\frac{-(\mu-1)\frac{8t}{3}-d}{\lambda} \le n \le \frac{-(\mu-1)\frac{8t}{3}+d}{\lambda}$$

$-61.11 \le n \le -27.7$

34 fringes are observed

18. (B)

By NLM

$$N\sin\theta = 4ma_4$$

$$3mg - N\cos\theta - T_1 = 3ma_2$$

$$T_1 - T_2\cos 60 = 2ma_2$$

$$T_2 - mg = ma_1$$

$$a_3 = a_4 = a_2$$

$$a_2\cos 60 = a_1$$

$$b = mg + \frac{5mg}{37} = \frac{42mg}{37}$$

$$3mg - N\cos\theta - \frac{mg}{2} = \frac{21ma}{4}$$

$$\frac{5mg}{2} = \frac{37ma}{4}$$

$$a = \left(\frac{10g}{37}\right)$$

So, $T_2 - \frac{mg}{2} = 2mg + \frac{mg}{2} = \frac{9ma_2}{4}$

PART-II (CHEMISTRY)

19. (9.48)

$NH_4OH + HCl \rightarrow NH_4Cl$

40	10	0
30	0	10

pOH = 5 + log $\frac{1}{3}$; pH = 9.48

20. (1.20)

$K_4[Fe(CN)_6] + 6H_2SO_4 + 6H_2O \xrightarrow{\Delta}$

$2K_2SO_4 + FeSO_4 + 3(NH_4)_2SO_4 + 6CO\uparrow$

$5CO + I_2O_5 \rightarrow 5CO_2 + I_2$

21. (6.00)

x = 2

y = 2

z = 2

So, (x + y + z) = 6

22. (1.00)

Only (vi) gives the test

23. (54.00)

Let molar mass of solute = M

$$\frac{p^\circ - 20}{p^\circ} = \frac{12/M}{(360/18)+(12/M)}$$

$$\frac{p^\circ - 20}{p^\circ} = \frac{12/M}{20+(12/M)} \quad \text{....(i)}$$

$$\frac{p^\circ - 20.02}{p^\circ} = \frac{12/M}{22+(12/M)} \quad \text{....(ii)}$$

Solving (i) and (ii)

p° = 20.22 torr

M = 54 g/mol

24. (0.48)

$$\underset{2-x}{N_2O_3(g)} \rightleftharpoons \underset{x-2y}{NO_2(g)} + \underset{x}{NO}$$

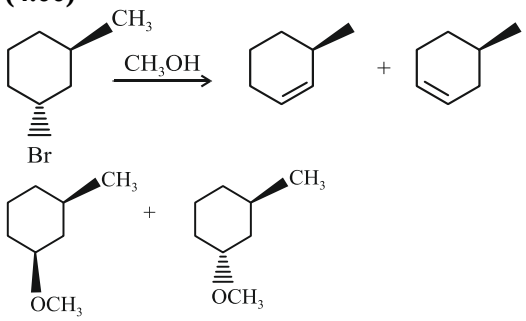
$$\underset{x-2y}{2NO_2(g)} \rightleftharpoons \underset{y}{N_2O_4(g)}$$

Given x = 1.5

$$2.5 = \frac{(1.5-2y)(1.5)}{0.5}$$

$$y = \frac{1}{3};\ K_{P_2} = \frac{(1/3)}{\left[1.5 - 2(1/3)\right]^2} = 0.48$$

25. (4.00)



26. (2.39)

$$\wedge^{\infty}_{AgCl} = \lambda_{Ag^+} + \lambda_{Cl^-}$$

$= 61.9 + 76.3$

$= 138.2 \text{ mho cm}^2 \text{ mol}^{-1}$

$$\frac{k \times 1000}{C} = \wedge^{\infty}_{m} = \frac{2.30 \times 10^{-6} \times 10^3}{138.2}$$

$= 1.66 \times 10^{-5} \text{ mol/L}$

$s = 1.66 \times 10^{-5} \times 143.5$

$= 2.39 \times 10^{-3} \text{ g/L}$

27. (A, B, D)

$$\Rightarrow K_f = \frac{M_1 R T_f^2}{\Delta_{fus} H_1}$$

$$= \frac{152 \times 10^{-3} \times 8.134 \times 450 \times 450}{8.53 \times 10^3}$$

$= 30 \text{ K kgmol}^{-1}$

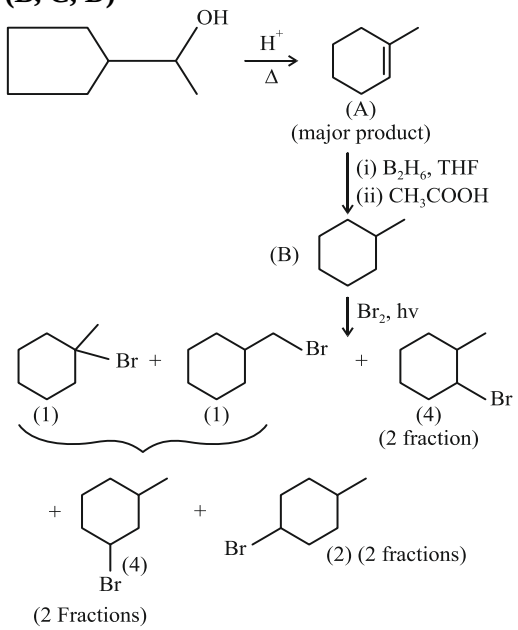
$\Rightarrow \Delta T_f = K_f.m$

$$m = \frac{450 - 430}{30} = \frac{20}{30} = 0.667$$

$$\Rightarrow m = \frac{\text{wt.}}{\text{M.wt}} \times \frac{1000}{\text{wt. of solvent}}$$

$$\text{M.wt.} = \frac{0.0386}{0.667} \times \frac{1000}{0.522} = 110.92$$

28. (B, C, D)



Total 12 mono-bromo products are formed.

29. (A, B, C, D)

With all the given anions, $AgNO_3$ gives white ppt.

30. (A, B, C)

Fact or information based

31. (A, B, C)

	$A(g) \rightarrow$	$2B(g)$ +	$C(g)$
t = 0	400	0	0
t = 20 min	400–P	2P	P

$P_{Total} = 400 + 2P = 1000$

$P = 300$

$$k = \frac{2.303}{20} \log \frac{400}{400 - 300} = \frac{2.303}{30} \log \frac{400}{400 - P}$$

$\Rightarrow P = 350 \text{ mm}$

32. (A, B)

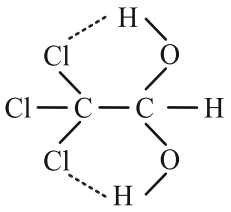
Reaction proceeds by limiting reagent only, Enthalpy of neutralization decreases, when acid or base or both are weak:

33. (A)

I. P, Q, R, S, T
II. P, R, S, T
III. Q, S, T
IV. Q, R, S, T

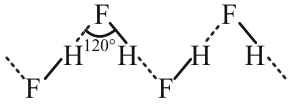
34. (A)

(I)



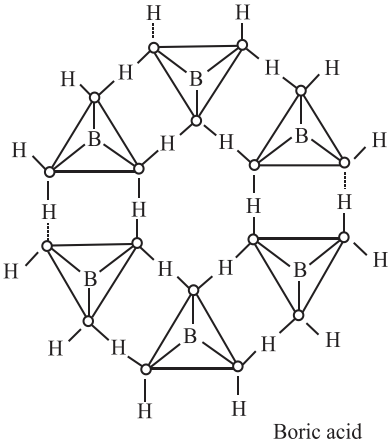
Chloral hydrate

(II)



Hydrogen fluoride

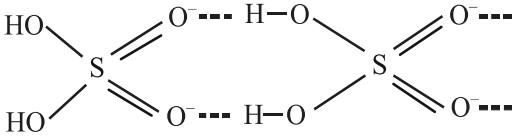
(III)



Boric acid

In the solid state, the $B(OH)_3$ units are hydrogen bonded together in two dimensional sheets with almost hexagonal

symmetry. The layers are quite a large distance apart (3.18 Å).

(IV) 

Sulphuric acid

35. (B)

I. (Q), II. (P, R, T), III. (P, S, T), IV. (Q, S, T)

(I) Mixing of ideal gas is spontaneous.

$W_{net} = 0$

$Q = 0$

$\therefore \quad \Delta U = 0$

$\Rightarrow \quad T \Rightarrow$ constant

$\Rightarrow \quad \Delta S_{system} > 0, \Delta S_{universe} > 0$

(II) $PV^2 = K \Rightarrow x = 2\,(PV^x = \text{ const})$

$$C_m = C_{v,m} + \frac{R}{1-x} = \frac{3}{2}R - R = \frac{R}{2}$$

$\Rightarrow$ TV = const

$\Rightarrow$ T decreases on increasing volume

$dQ = nC_m dT$

$\Rightarrow$ gas loses heat

Process is reversible $\therefore \Delta S_{universe} = 0$

(III) $PV^{\frac{3}{2}} = \text{const} \Rightarrow TV^{\frac{1}{2}} = \text{const}$

$$x = \frac{3}{2}$$

$$\therefore C_m = \frac{3}{2}R + \frac{R}{1-\frac{3}{2}} = -\frac{R}{2} \Rightarrow (-)Ve$$

On increasing V, T will decreases

$\therefore \quad dQ = nC_m dT \Rightarrow (+)$ Ve

$\therefore$ The gas gain heat.

Process is reversible $\therefore \Delta S_{universe} = 0$

(IV) $T = \dfrac{PV}{nR}$

$\Rightarrow$ T increases b/w 1 to 2 & work is done by the system

$\therefore$ gas must gain heat.

Process is reversible $\therefore \Delta S_{universe} = 0$

36. (D)

$I \rightarrow S$, $II \rightarrow P$, $III \rightarrow P,Q,S,T$, $IV \rightarrow R$

Refer isomerism in Co-ordination compounds.

PART-III (MATHEMATICS)

37. (7.00)

Comparing the coefficient of x^n both sides we get

$b_n - b_{n-1} = a_n$

Also, $b_0 = a_0$ and $b_1 = a_0 + a_1 \Rightarrow b_0 = 1, a_1 = 2$

Area bounded by $y = \dfrac{7}{1+x^2}$

x-axis, y-axis and $x = \sqrt{3}$ will be

$$\int_0^{\sqrt{3}} \frac{7}{1+x^2}\,dx = \frac{7\pi}{3}$$

38. (13.75)

$$\sum_{q=1}^{10}\sum_{r=1}^{10}\sum_{p=1}^{10} p\tan^{-1}\left(\frac{q}{r}\right)$$

$$= \sum_{q=1}^{10}\sum_{r=1}^{10}\tan^{-1}\left(\frac{q}{r}\right)\times\frac{10\times11}{2}$$

$$= 55.[25\pi] = 1375\,\pi$$

39. (5.20)

$$|z|_{max} = \sqrt{5^2+12^2}+1 = 14$$

$$|z|_{min} = \sqrt{5^2+12^2}-1 = 12$$

40. (0.00)

$$\text{If } \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

So, $p(x)$ is a constant function

$p(x) = 2$

41. (47.00)

Total words are $\dfrac{10!}{2!2!2!}$

The words contains the word "IIT" are $\dfrac{8!}{2!}$

$$\Rightarrow \text{Probability} = \frac{\frac{8!}{2!}}{\frac{10!}{2!2!2!}} = \frac{2}{45}$$

42. (164.00)

Class	Frequency	C.F
10-20	α	α
20-30	110	α + 110
30-40	54	α + 164
40-50	30	α + 194
50-60	β	α + β + 194 = 584

$$\overline{N = \sum f = 584}$$

$$\alpha + \beta = 390$$

$$\text{Median (m)} = \ell + \left[\frac{\left(\frac{N}{2}\right)-c}{f}\right]\times h$$

$$N = \frac{584}{2} = 292$$

$$m = 45 = 40 + \left[\frac{292 - (\alpha + 164)}{30}\right] \times 10$$

$$45 = 40 + \left(\frac{128 - \alpha}{3}\right)$$

$$5 = \frac{128 - \alpha}{3}$$

$15 = 128 - \alpha$

$\alpha = 113$

$\beta = 277$

$|\alpha - \beta| = |113 - 277| = 164$

43. (6.00)

$AB = 22 = AC$

Now $AP + PR + AR = (AP + PQ) + (QR + AR)$

$= AB + AC = 44.$

44. (1.50)

$$\sum_{k=1}^{\infty}\sum_{r=1}^{k}\frac{1}{3^k}\left({}^kC_r\right) = \sum_{k=1}^{\infty}\frac{1}{3^k}\sum_{r=1}^{k}{}^kC_r$$

$$= \sum_{k=1}^{\infty}\frac{1}{3^k}\cdot\left(2^k - 1\right)$$

$$= \sum_{k=1}^{\infty}\frac{2^k}{3^k} - \sum_{k=1}^{\infty}\frac{1}{3^k} = 2 - \frac{1}{2} = \frac{3}{2}$$

$$\therefore \quad \left(\frac{2}{3} + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \ldots\ldots\infty = \frac{\frac{2}{3}}{1 - \frac{2}{3}} = 2\right)$$

45. (B, C)

(A) $f(x) = \{x\}\cos^2\left(\left(\frac{2x-1}{2}\right)\pi\right)$ differentiable

$\forall\ x \in R$

(B) not differentiable at $x = (2n + 1)\ \pi$

(C) not differentiable at $x = 0, 2, 6, 4$

(D) not differentiable at all integers

46. (B, D)

For ellipse: $e = \sqrt{\frac{9-4}{9}} = \frac{\sqrt{5}}{3}$

Foci are $\left(\pm\sqrt{5}, 0\right)$

For hyperbola: $ae = \sqrt{5}$, $e = 2$, $a = \frac{\sqrt{5}}{2}$,

$$b = \frac{\sqrt{15}}{2}$$

$L.R = 3\sqrt{5}$, $a^2 + b^2 = 5$

47. (B, C)

$a = 2$

$b = \sqrt{3}$

$PS' = \frac{4}{3}$

48. (C, D)

$p(x) = ax^2 + (a-1)x + (a-2)$ has root α and β, then $|\alpha - \beta| < 2$

$$\Rightarrow \sqrt{\left(-\left(\frac{a-1}{a}\right)\right)^2 - 4\left(\frac{a-2}{a}\right)} < 2$$

$$\Rightarrow \frac{-7a^2 + 6a + 1}{a^2} < 0$$

$$\Rightarrow a \in \left(-\infty, -\frac{1}{7}\right) \cup (1, \infty) \text{ and discriminant } D > 0$$

$$\Rightarrow (a-1)^2 - 4a(a-2) > 0$$

$$\Rightarrow a \in \left(1 - \frac{2}{\sqrt{3}}, 0\right) \cup \left(0, 1 + \frac{2}{\sqrt{3}}\right)$$

So, the answer is $\left(1 - \frac{2}{\sqrt{3}}, -\frac{1}{7}\right) \cup \left(1, 1 + \frac{2}{\sqrt{3}}\right)$

49. (A, D)

Since $\int_a^b f(x)\,dx = (b-a)\int_0^1 f\{(b-a)x + a\}\,dx$,

$$\int_1^2 \sin x^2 dx = \int_0^1 \sin(x+1)^2\,dx = \int_0^1 \sin\left(x^2 + 2x + 1\right)dx$$

$$\int_{-4}^{4} \cos x^2 dx = 8\int_0^1 \cos(8x-4)^2\,dx$$

$$= 8\int_0^1 \cos 16(2x-1)^2\,dx$$

50. (A, B, D)

$$\frac{dy}{dx}\left(x^2y^3 + xy\right) = 1$$

$$\Rightarrow \quad \frac{dx}{dy} = x^2y^3 + xy$$

$$\Rightarrow \quad \frac{dx}{dy} - xy = x^2y^3$$

$$\Rightarrow \quad -\frac{1}{x^2}\frac{dx}{dy} + \frac{y}{x} = -y^3$$

Put, $\frac{1}{x} = t$ and differentiating with respect to y

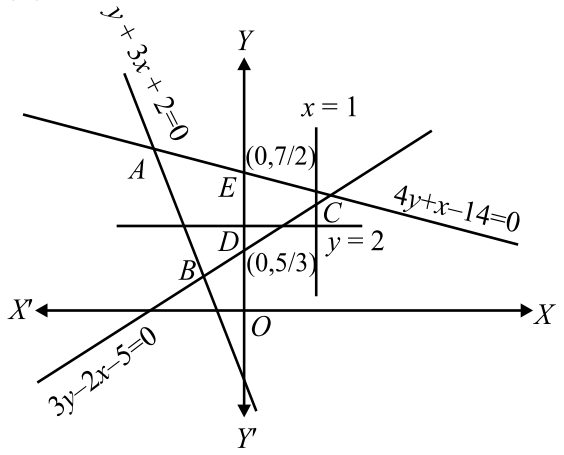
$$-\frac{1}{x^2}\frac{dx}{dy} = \frac{dt}{dy}$$

$$\Rightarrow \quad \frac{dt}{dy} + ty = -y^3$$

$$\Rightarrow \quad e^{\frac{y^2}{2}} t = \int -y^3 e^{y^2/2} dy$$

$$\Rightarrow \quad \frac{1}{x} = 2 - y^2 + c\, e^{-\frac{y^2}{2}}$$

51. (C)



(I) The points on the line $x = 0$, whose y-coordinate lies between $\frac{5}{3}$ and $\frac{7}{2}$ inside the triangle ABC.

$\therefore \quad \frac{5}{3} < \lambda < \frac{7}{2}$ or $5 < 3\lambda < 10.5$

$\therefore \quad |3\lambda| = 6, 7, 8, 9, 10$

(II) $\because \quad C \equiv (2, 3)$

The points on the line $x = 1$, whose y-coordinate lies between

$\frac{7}{3}$ (put $x = 1$ in $3y - 2x - 5 = 0$)

and $\frac{13}{4}$ (put $x = 1$ in $4y + x - 14 = 0$)

$\therefore \quad \frac{7}{3} < \lambda < \frac{13}{4}$ or $7 < 3\lambda < 9.75$

$\therefore \quad |3\lambda| = 8, 9$

(III) $\because \quad B \equiv (-1, 1)$

The point on the line $y = 2$, whose x-coordinate lies between

$-\frac{4}{3}$ (put $y = 2$ in $y + 3x + 2 = 0$)

And $\frac{1}{2}$ (put $y = 2$ in $3y - 2x - 5 = 0$)

$\therefore \quad -\frac{4}{3} < \lambda < \frac{1}{2}$ or $-8 < 6\lambda < 3$

Integral values of 6λ are

$-7, -6, -5, -4, -3, -2, -1, 0, 1, 2$

$\therefore \quad |6\lambda| = 7, 6, 5, 4, 3, 2, 1, 0$

(IV) $\because \quad A \equiv (-2, 4)$

The points on the line $y = \frac{7}{2}$, whose x-coordinates lies between

0 (put $y = \frac{7}{2}$ in $4y + x - 14 = 0$)

and $-\frac{11}{6}$ (put $y = \frac{7}{2}$ in $y + 3x + 2 = 0$)

$\therefore \quad -\frac{11}{6} < \lambda < 0$

or $\quad -11 < 6\lambda < 0$

Integral value of 6λ are

$-10, -9, -8, -7, -6, -3, -2, -1$

$\therefore \quad |6\lambda| = 10, 9, 8, 7, 6, 5, 4, 3, 2, 1$

52. (A)

(I) $\frac{3}{5} \geq P(A \cap B) \geq P(A) + P(B) - 1$

$= \frac{3}{5} + \frac{2}{3} - 1 = \frac{4}{15}$

$\therefore \quad \frac{3}{5} \geq P(A \cap B) \geq \frac{4}{15}$

$\Rightarrow \quad P(A \cap B) \in \left[\frac{4}{15}, \frac{3}{5}\right]$

(II) $\because \; P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{3}{5} + \frac{2}{3} - P(A \cap B) = \frac{19}{15} - P(A \cap B)$

$\Rightarrow \quad \frac{2}{3} \leq P(A \cup B) \leq 1$

$\left[\because \; P(A \cap B) \in \left[\frac{4}{15}, \frac{3}{5}\right]\right]$

$\therefore \quad P(A \cup B) \in \left[\frac{2}{3}, 1\right]$

(III) $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{3}{2} P(A \cap B)$

$\Rightarrow \quad \frac{2}{5} \leq P\left(\frac{A}{B}\right) \leq \frac{9}{10} \left[\because P(A \cap B) \in \left[\frac{4}{15}, \frac{3}{5}\right]\right]$

$\therefore \quad P\left(\frac{A}{B}\right) \in \left[\frac{2}{5}, \frac{9}{10}\right]$

(IV) $P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)} = \frac{5}{3} P(A \cap B)$

$\Rightarrow \quad \frac{4}{9} \leq 9\left(\frac{B}{A}\right) \leq 1$

$\left[\because \; P(A \cap B) \in \left[\frac{4}{15}, \frac{3}{5}\right]\right]$

$\therefore \quad P\left(\frac{B}{A}\right) \in \left[\frac{4}{9}, 1\right]$

53. (D)

$12 \leq n(A \cup B) \leq 22$

And $0 \leq n(A \cap B) \leq 10$

54. (D)

$L_1 : \frac{x}{3} = \frac{y-3}{4} = \frac{z+2}{1}$ …(i)

(passing through P and Q)

$L_2 : \frac{x-1}{1} = \frac{y+3}{0} = \frac{z+1}{1}$ ….(ii)

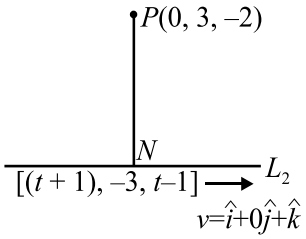
(Passing through R and parallel to $v = \hat{i} + \hat{k}$)

(I) Distance of $P(0, 3, -2)$ from L_2

$PN = (t+1)\hat{i} - 6\hat{j} + (t+1)\hat{k}$

Now, $PN \cdot V = 0$

$\left[(t+1)\hat{i} - 6\hat{j} + (t+1)\hat{k}\right] \cdot \left(\hat{i} + \hat{k}\right) = 0$



$(t + 1) + (t + 1) = 0; t = -1$

$N(0, -3, -2)$

$PN = \sqrt{0^2 + 6^2 + 0^2} = 6$

(II) Distance between L_1 and L_2

Equation of plane containing L_1 and parallel to L_2

$Ax + B(y - 3) + C(z + 2) = 0$

Where $3A + 4B + C = 0$

And $A + B + C = 0$

$A + C = 0$

$C = \lambda, A = -\lambda, B = +\lambda/2$

$\therefore$ Equation of plane

$-\lambda x + \frac{\lambda}{2}(y-3) + \lambda(z+2) = 0$

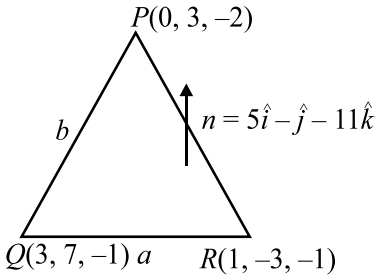
$2x - y + 3 - 2z - 4 = 0$

$2x - y - 2z = 1$ …(i)

Now, distance of the point (1, –3, –1) lying on the line L_1 from the plane (i)

$d = \left|\frac{2+3+2-1}{3}\right| = 2$

(III) Area of ΔPQR



$QR = a = 2\hat{i} + 10\hat{j} + 0\hat{k}$

$QP = b = 3\hat{i} + 4\hat{j} + \hat{k}$

$a \times b = 2\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 5 & 0 \\ 3 & 4 & 1 \end{vmatrix}$

$= 2\left[\hat{i}(5) - \hat{j}(1) + \hat{k}(4-15)\right] = 2\left[5\hat{i} - \hat{j} - 11\hat{k}\right]$

$\frac{|a \times b|}{2} = \sqrt{25+1+121} = \sqrt{147}$

$= \sqrt{3 \cdot 49} = 7\sqrt{3}$

(IV) Distance of P from centroid (0, 0, 0) of ΔQRS is $\sqrt{13}$



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